

Biochar - Certificate of Analysis - E26-00-1894

Client:	Onetrak Pty Ltd	Laboratory:	Environmental Analysis Laboratory
Contact:	Sean Curtis	Contact:	EAL Customer Service Team
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Customer reference:	Green Waste	Request ID:	EAL /E26-00-1894
Number of samples:	1	Report ID:	E26-00-1894_EALS1_1
Date samples received:	25 February 2026	Issue date:	15 March 2026

Authorised by:	Brian Smith
Position:	Senior Technical Officer



Comments: EAL is a NATA accredited laboratory (14960), accredited for compliance with ISO/IEC 17025 - Testing.

Sample date received: 20/02/2026

* raised LOR due to matrix interference for PAH results.

Certificate of Analysis

Request ID: EAL/E26-00-1894 Report ID: E26-00-1894_EALS1_1 Issue date: 15 March 2026

Client Sample ID:				Green Waste Biochar
EAL Sample ID:				E26-00-1894-0001
Parameter	Unit	Method Reference	LOR	---
Wet Bulk Density	kg/L	** AS4454:2012 Appendix J	---	0.58
Dry Bulk Density	kg/L	** AS4454:2012 Appendix J	---	0.28
Moisture Content 105°C	%	** AS4454:2012 Appendix I	<0.1	52
pH (1:10 H ₂ O)	---	** Inhouse S95	---	10.1
EC (1:10 H ₂ O)	dS/m	** Inhouse S95	---	1.96
Loss on Ignition	%	** Inhouse S91	---	68.1
Ash Content	%	** Inhouse S91	---	31.9
Hydrogen	%	** de Groot 2009	<0.01	1.60
Oxygen	%	** de Groot 2009	<0.01	4.09
Hydrogen/TOC Molar Ratio	---	** de Groot 2009	<0.01	0.31
Oxygen/TOC Molar Ratio	---	** de Groot 2009	<0.01	0.05
Total Organic Carbon	%	Inhouse S15b	<0.10	62.2
Total Carbon	%	Inhouse S4a	<0.03	66.6
Total Nitrogen	%	Inhouse S4a	<0.02	0.19
Neutralising Value	% CaCO ₃	** Inhouse S14	<0.01	5.47
Sulfur	%	** ICPOES Total S - Compost	<0.01	0.07
Calcium	%	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.01	2.59
Magnesium	%	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.01	0.49
Potassium	%	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.01	1.25
Sodium	%	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.01	0.31
Phosphorus	%	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.01	0.98
Zinc	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	277
Manganese	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	523
Iron	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<50	14400
Copper	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	143
Boron	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<2	28.6
Silicon	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<50	1510
Aluminium	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<50	14200
Molybdenum	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.2	2.50
Cobalt	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	3.53
Selenium	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.5	1.82
Cadmium	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.5	< 0.5
Lead	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	22.3
Arsenic	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<2	13.1
Chromium	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<2	27.3
Nickel	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	9.36
Mercury	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<0.1	0.28
Silver	mg/kg	** 1:3 Nitric:HCl - APHA 3125 ICPMS	<1	< 1
Particle and Surface Sciences Report No.	---	**Subcontracted PSS	---	PsS Testing
Single Point Surface Area P/PO	m ² /g	**Subcontracted PSS	---	366.4337
BET Surface Area	m ² /g	**Subcontracted PSS	---	369.4341

Certificate of Analysis

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Client Sample ID:				Green Waste Biochar
EAL Sample ID:				E26-00-1894-0001
Parameter	Unit	Method Reference	LOR	---
Langmuir Surface Area	m ² /g	** Subcontracted PSS	---	414.4257
SGS Report No.	---	** Subcontracted SGS Laboratories Australia	---	<1.6
Arochlor 1016	mg/kg	** Subcontracted SGS Laboratories Australia	---	<1.6
Arochlor 1221	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
Arochlor 1232	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
Arochlor 1242	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
Arochlor 1248	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
Arochlor 1254	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
Arochlor 1260	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
Total PCBs (Arochors)	mg/kg	** Subcontracted SGS Laboratories Australia	---	<0.2
SGS Report No.	---	** Subcontracted SGS Laboratories Australia	---	SE298128
Napthalene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
2-methylnaphthalene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
1-methylnaphthalene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Acenaphthalene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Acenaphthene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Fluorene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Phenathrene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Anthracene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Fluoranthene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Pyrene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Benzo(a)anthracene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Benzo(k)fluoranthene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Chrysene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Benzo(b&j)fluoranthene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Benzo(a)pyrene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Indeno(1,2,3-cd)pyrene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Dibenzo(ah)anthracene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Benzo(ghi)perylene	mg/kg	** Subcontracted SGS Laboratories Australia	<0.1	<0.2
Carcinogenic PAHs, BaP TEQ <LOR=0*	TEQ (mg/kg)	** Subcontracted SGS Laboratories Australia	<0.2	<0.4
Carcinogenic PAHs, BaP TEQ <LOR=LOR*	TEQ (mg/kg)	** Subcontracted SGS Laboratories Australia	<0.3	<0.6
Carcinogenic PAHs, BaP TEQ <LOR=LOR/2*	TEQ (mg/kg)	** Subcontracted SGS Laboratories Australia	<0.2	<0.4
Total PAH (18)	mg/kg	** Subcontracted SGS Laboratories Australia	<0.8	<1.6
Total PAH (NEPM/WHO 16)	mg/kg	** Subcontracted SGS Laboratories Australia	<0.8	<1.6

Certificate of Analysis

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Notes:

- Total Acid Extractable Nutrients indicate a store of nutrients.
- < 'LOR' means less than limit of reporting. 'ND' means not detected. 'NA' means not applicable. 'NR' means not requested.
- Analysis completed in a NATA accredited laboratory.
- ** denotes NATA accreditation does not cover the performance of this service.
- .. denotes not requested, no data/information or no guidelines available.
- All services undertaken by EAL are covered by the EAL Laboratory Services Terms and Conditions (available on request or at scu.edu.au/eal).
- Analysis conducted between sample arrival date and reporting date.
- This report is not to be reproduced except in full.
- Results only relate to the item tested.
- Methods from Rayment and Lyons, 2011. Soil Chemical Methods - Australasia. CSIRO Publishing, Collingwood.
- All analysis is on a dry weight (DW) basis - testing conducted on finely ground sample dried at 40 °C, with results corrected to 105 °C.
- Analysis reported on a dry weight (DW) basis, unless wet weight (WW) specified.
- Conversions for 1 mg/kg = 1 ppm 1 % = 10,000 ppm.



TriStar 3000 V6.08 A

Unit 1 Port 2

Serial #: 1586

Page 1

Sample: E26 01 1894 0001
Operator: MG
Submitter: Southern Cross University EAL
File: C:\...\TRISTAR\001-372.SMP

Started: 12/03/2026 12:36:36PM
Completed: 13/03/2026 3:56:27PM
Report Time: 13/03/2026 4:24:23PM
Warm Free Space: 8.6482 cm³ Measured
Equilibration Interval: 25 s
Sample Density: 1.000 g/cm³

Analysis Adsorptive: N2
Analysis Bath Temp.: 77.300 K
Sample Mass: 0.3920 g
Cold Free Space: 30.7656 cm³ Measured
Low Pressure Dose: None
Automatic Degas: No

Isotherm Tabular Report

Relative Pressure (P/Po)	Absolute Pressure (mmHg)	Quantity Adsorbed (cm ³ /g STP)	Elapsed Time (h:min)	Saturation Pressure (mmHg)
			04:25	762.53241
			06:26	765.40778
			08:26	766.88739
			10:26	768.82220
			12:26	769.48712
			14:26	769.89447
0.024682508	19.02239	85.5236	16:23	
			16:26	770.70319
0.030618519	23.62563	86.5666	17:52	
			18:25	771.96112
0.048473436	37.42885	88.4639	19:01	
0.104341658	80.60851	91.3818	20:15	
			20:26	772.60211
0.165030838	127.50317	93.0890	20:45	
0.209506749	161.86536	94.0099	21:06	
0.260444263	201.21979	94.7601	21:13	
0.304567737	235.30968	95.1835	21:19	

TriStar 3000 V6.08 A

Unit 1 Port 2

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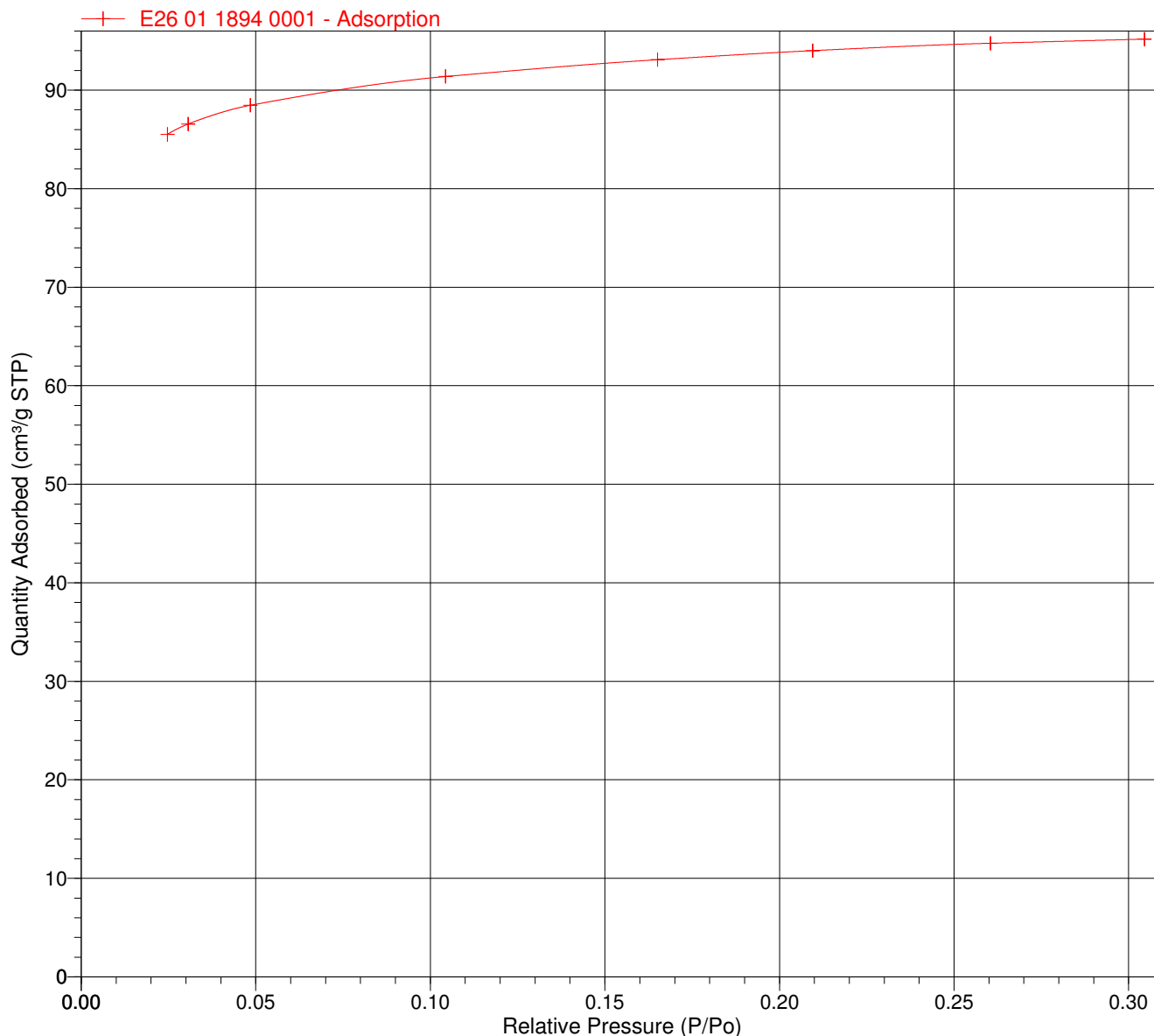
Page 2

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Low Pressure Dose: None
Automatic Degas: No

Isotherm Linear Plot



TriStar 3000 V6.08 A

Unit 1 Port 2

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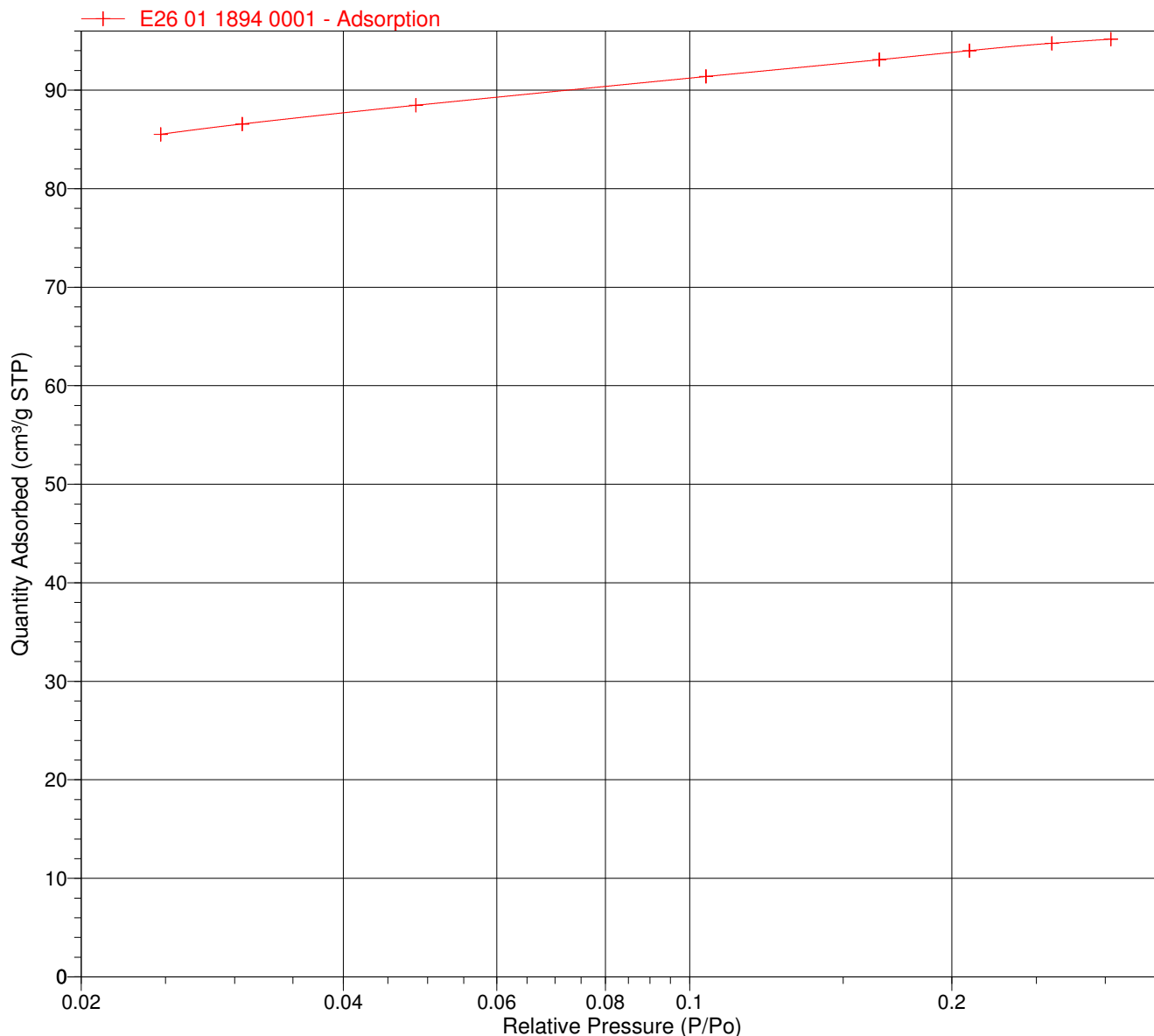
Page 3

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Analysis Bath Temp.: 77.300 K
Sample Mass: 0.3920 g
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Low Pressure Dose: None
Automatic Degas: No

Isotherm Log Plot



TriStar 3000 V6.08 A

Unit 1 Port 2

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Page 4

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Cold Free Space: 30.7656 cm³ Measured
Low Pressure Dose: None
Automatic Degas: No

BET Surface Area Report

BET Surface Area: 369.4341 ± 1.2471 m²/g
Slope: 0.011779 ± 0.000040 g/cm³ STP
Y-Intercept: 0.000005 ± 0.000001 g/cm³ STP
C: 2469.341727
Qm: 84.8650 cm³/g STP
Correlation Coefficient: 0.9999943
Molecular Cross-Sectional Area: 0.1620 nm²

Relative Pressure (P/Po)	Quantity Adsorbed (cm ³ /g STP)	1/[Q(Po/P - 1)]
0.024682508	85.5236	0.000296
0.030618519	86.5666	0.000365
0.048473436	88.4639	0.000576

TriStar 3000 V6.08 A

Unit 1 Port 2

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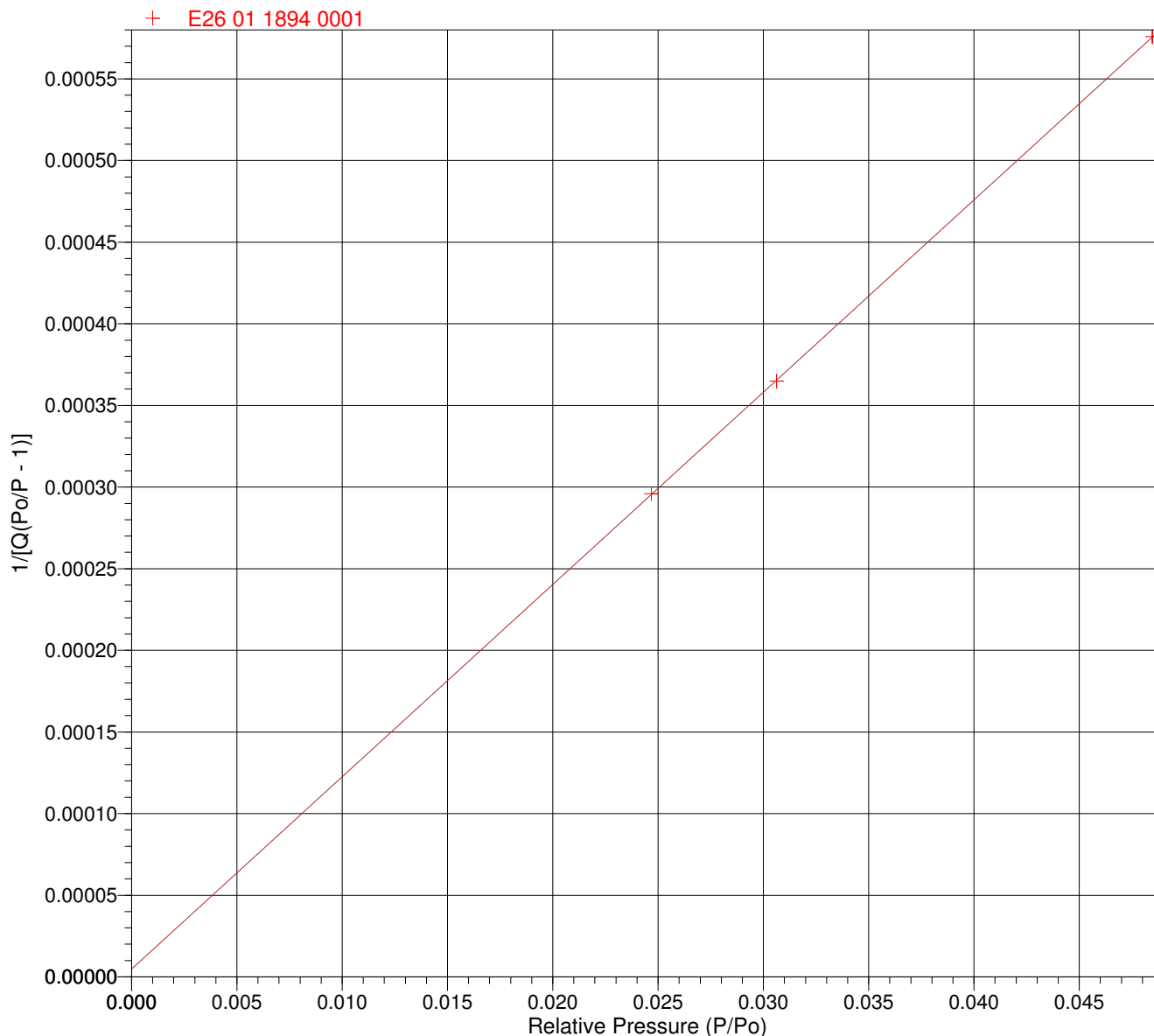
Page 5

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Analysis Bath Temp.: 77.300 K
Sample Mass: 0.3920 g
Cold Free Space: 30.7656 cm³ Measured
Low Pressure Dose: None
Automatic Degas: No

BET Surface Area Plot



TriStar 3000 V6.08 A

Unit 1 Port 2

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Analysis Bath Temp.: 77.300 K
Sample Mass: 0.3920 g
Cold Free Space: 30.7656 cm³ Measured
Low Pressure Dose: None
Automatic Degas: No

Langmuir Surface Area Report

Langmuir Surface Area: 414.4257 ± 1.7850 m²/g
Slope: 0.010504 ± 0.000045 g/cm³ STP
Y-Intercept: 0.027431 ± 0.004182 mmHg·g/cm³ STP
b: 0.382936 1/mmHg
Qm: 95.2003 cm³/g STP
Correlation Coefficient: 0.999963
Molecular Cross-Sectional Area: 0.1620 nm²

Pressure (mmHg)	Quantity Adsorbed (cm ³ /g STP)	P/Q (mmHg·g/cm ³ STP)
19.02239	85.5236	0.222
23.62563	86.5666	0.273
37.42885	88.4639	0.423
80.60851	91.3818	0.882
127.50317	93.0890	1.370
161.86536	94.0099	1.722

TriStar 3000 V6.08 A

Unit 1 Port 2

Serial #: 1586

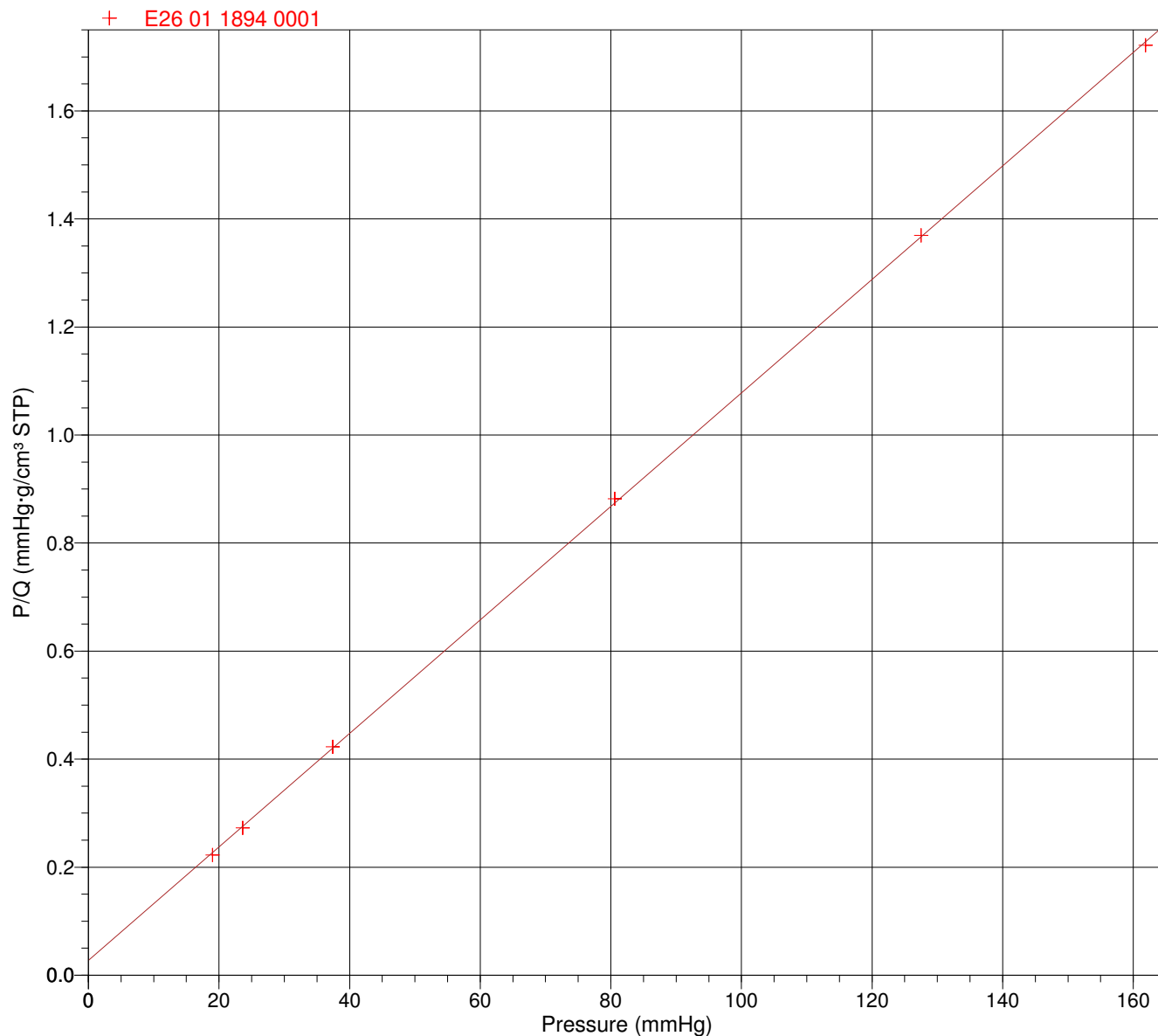
Page 7

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Analysis Adsorptive: N2
Analysis Bath Temp.: 77.300 K
Sample Mass: 0.3920 g
Cold Free Space: 30.7656 cm³ Measured
Low Pressure Dose: None
Automatic Degas: No

Langmuir Surface Area Plot





TriStar 3000 V6.08 A

Unit 1 Port 2

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Page 8

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Analysis Bath Temp.: 77.300 K
Sample Mass: 0.3920 g
Cold Free Space: 30.7656 cm³ Measured
Low Pressure Dose: None
Automatic Degas: No

Summary Report

Surface Area

Single point surface area at $P/P_0 = 0.048473436$: 366.4337 m²/g

BET Surface Area: 369.4341 m²/g

Langmuir Surface Area: 414.4257 m²/g